Love of Variety Model and Simulation

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1 Model

1.1 Consumer's Problem

The consumer's problem is to maximize utility subject to their choice set. The utility function is as follows:

$$U_{ijt} = \beta_i X_{jt} - \gamma_i (\Sigma_{ijt})^2 + \kappa a_{jt} - \alpha p_{jt} + \xi_j + \epsilon_{ijt} \tag{1}$$

$$\Sigma_{ijt} = X_{jt} - \bar{X}_{it}$$

Here, U_{ijt} is the utility of consumer i who chooses product j in period t. Utility depends on utility from characteristics associated with product X_jt, β_i, love of variety γ_i, how different your current choice is from the mean Σ_{ijt}, advertising a_{jt}, prices p_{jt}, and unobserved quality ξ_j. Further, a gumbel shock ε_{ijt} also impacts utility. This is what allows us to construct choice probabilities as follows:

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}} \tag{2}$$

2 Simulation

To understand model behavior, I first do Monte Carlo exercises across different environments. This will be replaced by Nielsen scanner data once access is gained.

2.1 Parameters

Parameter	Value	Description
β_i	$\{1, 2.5, 4\}$	Consumer preference for product attributes
γ_i	$\{1, 2.5, 4\}$	Consumer love of variety
κ	3	Advertising cost parameter
α_j	0	Price sensitivity
ξ_j	0	Unobserved quality

Table 1: Simulation Parameters

Parameters were informed by the gumbel distribution for scaling. Prices and unobserved quality are set to 0 for simplicity in early phases of testing. Will be added as I get further into the project.

2.2 Simulation Procedure

I have simulated two specifications so far: one in which the mean is fixed as the mean of an ordinal product set and one with a mean informed by "prior preferences" over products from a normal distribution. Below I go into each of these with graphs of choice probabilities in a market with one consumer.

2.3 Set Mean

In the first simulation routine, The product set J is defined as $J = \{1.0, 2.0, 3.0, 4.0, 5.0\}$. Consider this the most simple form available, where $\bar{X}_i t = 2.5$, the mean of J. Thus, $\Sigma_{ijt} = X_{jt} - 2.5$. Therefore, the only thing changing over time is the new epsilon draw each period. This specification also sets $a_{jt} = 0$ to really allow the β_i, γ_i regimes to show their behavior.

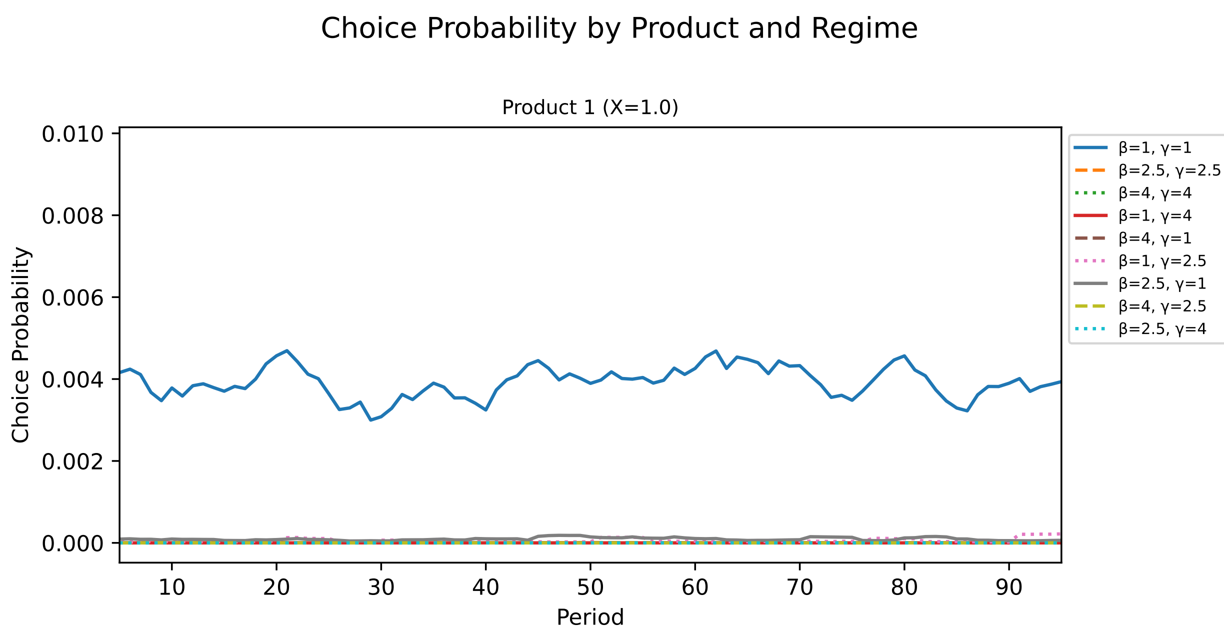

Figure 1: Choice Probability of Product 1 by Regime

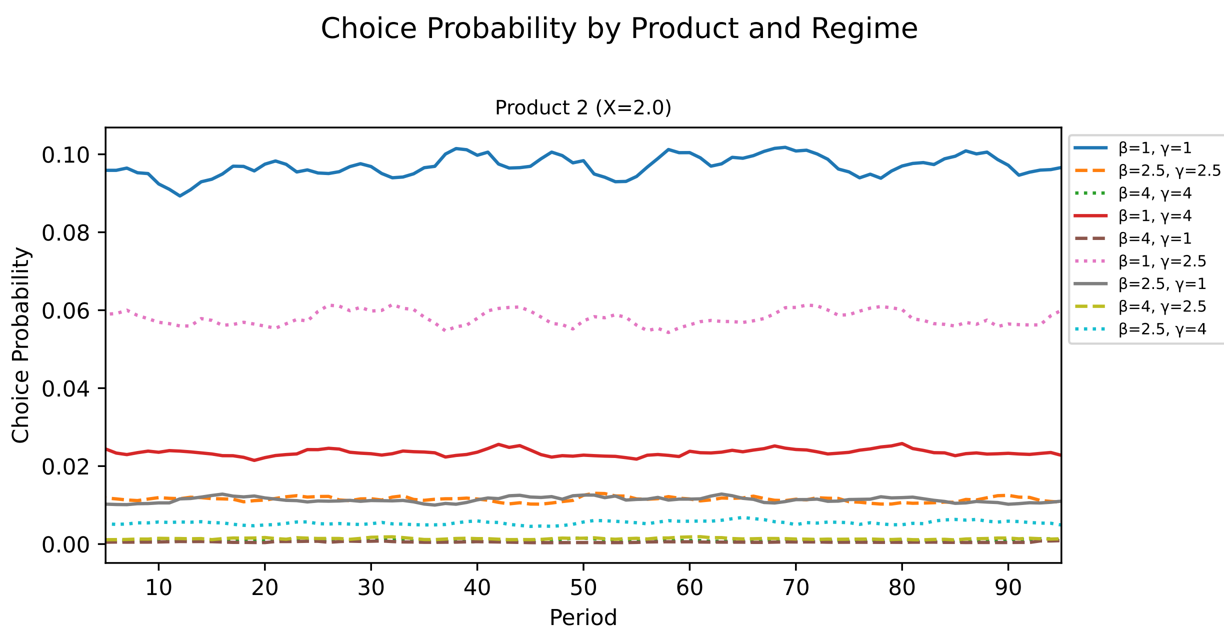

Figure 2: Choice Probability of Product 2 by Regime

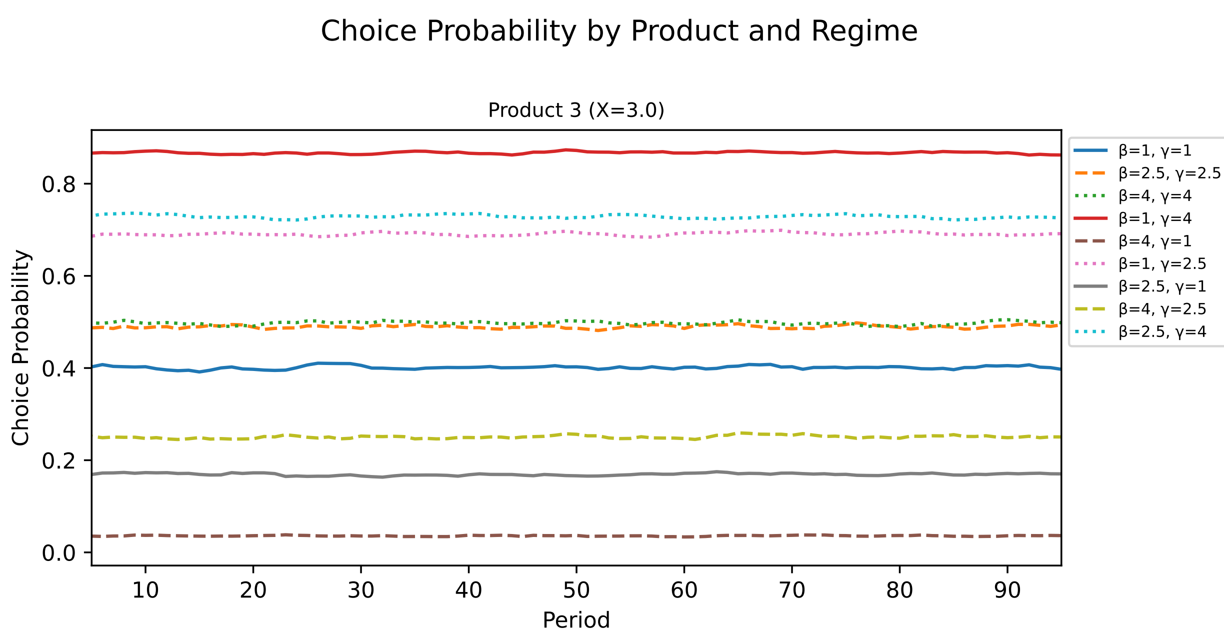

Figure 3: Choice Probability of Product 3 by Regime

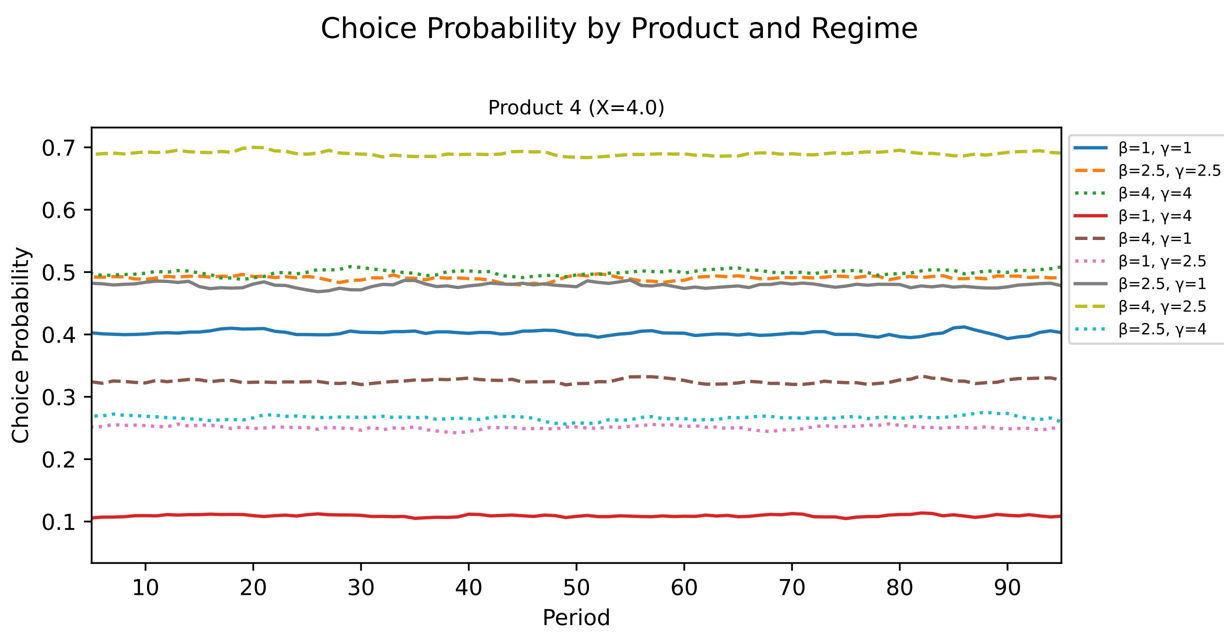

Figure 4: Choice Probability of Product 4 by Regime

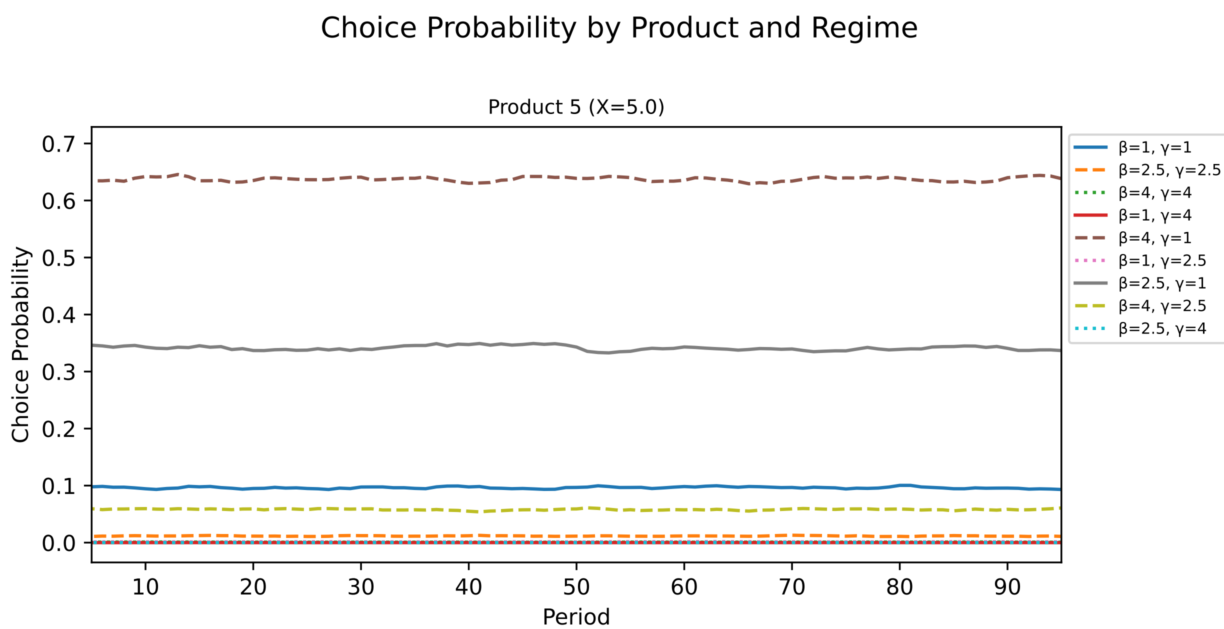

Figure 5: Choice Probability of Product 5 by Regime (Fixed Advertising $a = 0.2$)

In this setup, choice probabilities are practically flat, as is expected. I will pick out a few lines. In product 3, the ($\beta = 1,\ \gamma = 4$) regime has the the highest probability of choosing this product. This is to be expected, as they want to minimize the γ term by getting as close to the mean as possible. For product 4, the ($\beta = 4,\ \gamma = 2.5$) has the highest probability. This is because their utility gain outweighs the penalty enough to choose a higher utility product, though not enough to choose the highest utility product. Product 5, is chosen most by the ($\beta = 4,\ \gamma = 1$) regime for similar reasons as descibed above. The low penalty allows them to "indulge" in a higher utility product.

2.4 Prior Mean

In this routine, I implemented an idea Josh had in a meeting: the consumer has priors over what they like. So, to implement a version of this, I sim five periods before my main simulation of 100 periods to create a mean over X's as they enter the model. Further, I simulated it with $J \sim N(0,1)$ with 5 products. I did this because in the case from the above section, regimes went to $X = \{4, 5\}$. I wanted to do this to test how I have things specified. Below are choice probabilities for the set advertising case ($\bar{2}.0$) and the evolving advertising case ($\bar{X}_i t - X_{jt-1}$).

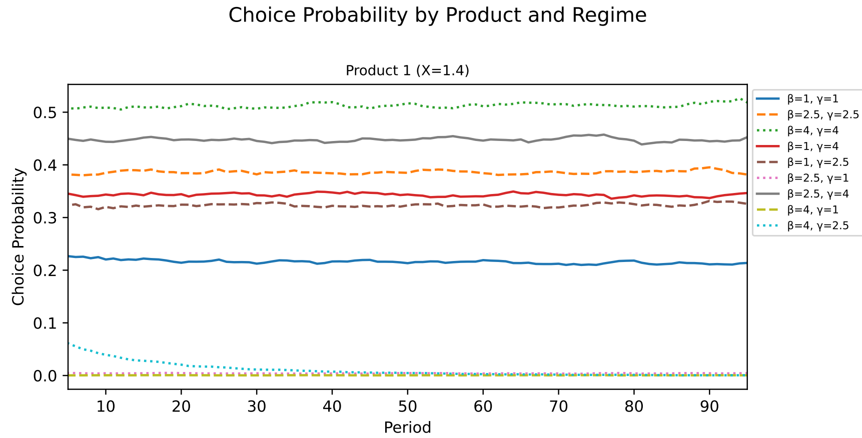

Figure 6: Choice Probability of Product 1 by Regime (Fixed Advertising $a = 2$)

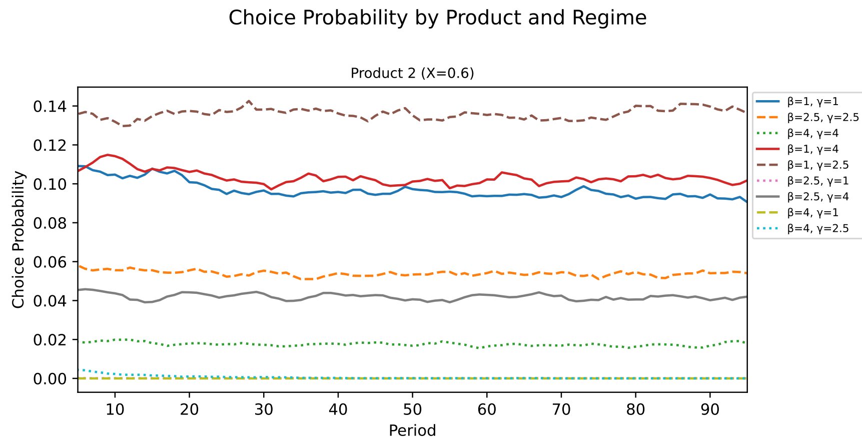

Figure 7: Choice Probability of Product 2 by Regime (Fixed Advertising $a = 2$)

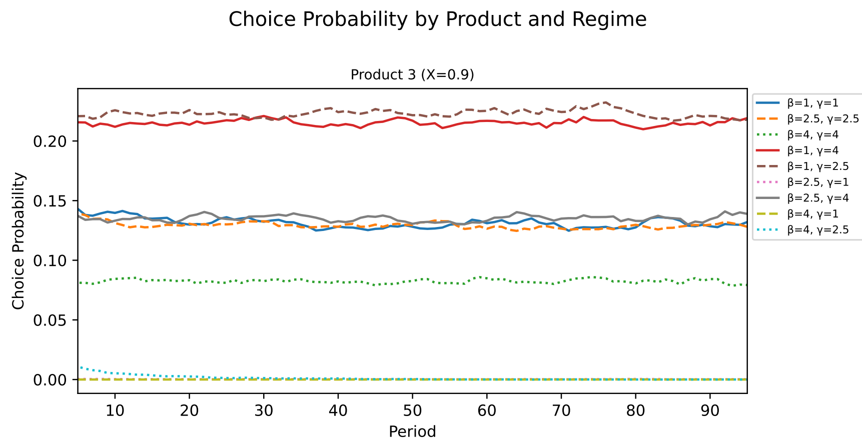

Figure 8: Choice Probability of Product 3 by Regime (Fixed Advertising $a = 2$)

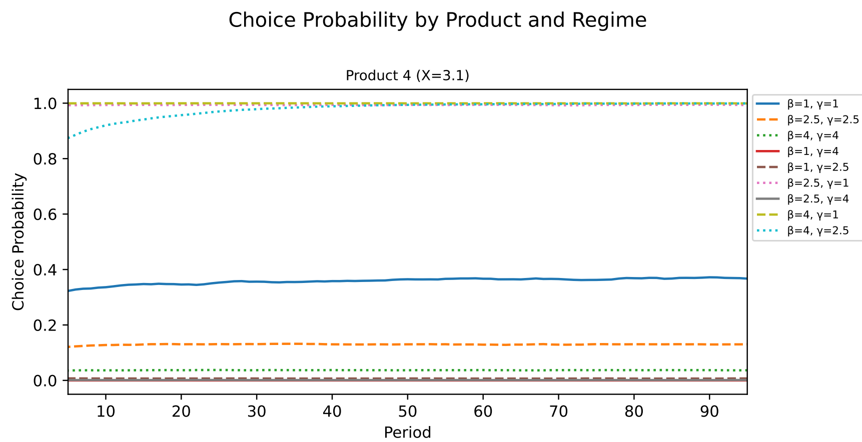

Figure 9: Choice Probability of Product 4 by Regime (Fixed Advertising $a = 2$)

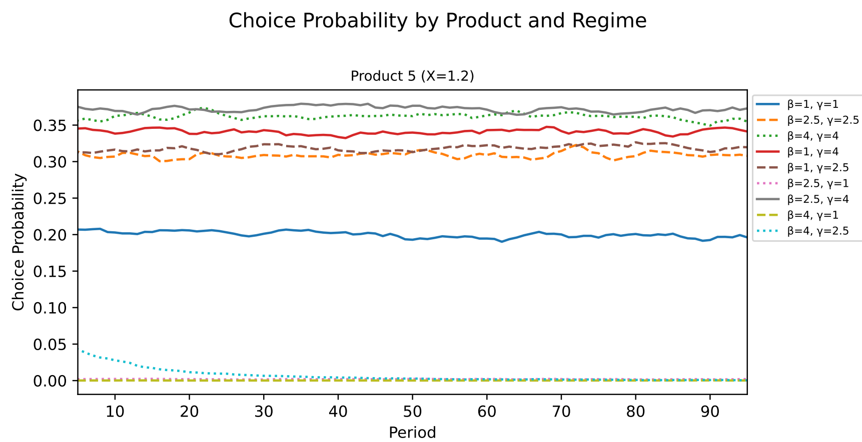

Figure 10: Choice Probability of Product 5 by Regime (Fixed Advertising $a = 2$)

This specification yields interesting patterns. Particularly, the regime ($\beta = 1, \gamma = 2.5$). This regime will flip between products, choosing all but product 4. This is because the utility gain is outweighed enough by the penalty to encourage variety. In the other specification, the same regime choosed product 3 70% of the time. So having a dynamic mean informed by priors seems to encourage the behaviors sought after. The regime ($\beta = 4, \gamma = 1$) picks the highest value product 100% of the time. ($\beta = 4, \gamma = 2.5$) converges to product 4 100% of the time after about 30 periods, showing that the penalty does have an effect early, but the utility gain is enough to eventually win out into steady state.

3 Next Steps

Next, I would like to go to 5 consumers, and see how things change. I also want to see how pulling from $\gamma \sim N(0, \sigma_\gamma)$ affects things. Of course, I also need to start really thinking about the firm side and the consumer's value function. I am also working on beginning my literature review, which has inspired some interesting ideas.